

voi-rank v1: ranking scenarios by the decision value of a single measurement

voi-rank pipeline (protocol p001, run 3)

August 21, 2026

Abstract

We rank 8 scenarios by the economic value of one measurement sold to one decision-maker facing one binary decision. Each scenario is scored by the expected value of sample information (EVSI) of the measurement relative to its production cost, computed in closed form from seven parameters elicited by a small language model under a fixed, anchored protocol, with elicitation uncertainty propagated by Monte Carlo (100 000 draws). The primary metric is *efficiency* = EVSI/C : decision value per dollar of measurement. Because buyer counts and value capture are deliberately out of scope, the ranking measures per-measurement decision-value density, not business feasibility.

1 Introduction

A measurement is a product: someone with a decision to make pays for one observation because they expect to decide better with it than without it. This report treats that framing literally. The unit of analysis is a single *scenario*: one concrete decision-maker (an agent), one binary decision, one latent binary fact about the world, and one purchasable instrument that emits one binary signal about that fact. The canonical Bayesian decision problem then prices the information: how much better, in expected USD, does the agent do with the signal than without it?

That price is the expected value of sample information (EVSI). Dividing by the cost C of producing and delivering the measurement gives an efficiency: how many dollars of decision value one dollar of measurement buys. Ranking many scenarios by this efficiency is a way of asking: *where in the economy is a single observation worth the most, per dollar it costs to produce?*

Two design choices keep v1 honest and small. First, everything that would turn information value into business value — how many buyers exist, what fraction of the value a seller can capture, recurrence, rivalry — is deliberately out of scope; the interpretation caveat in Section 7 is load-bearing. Second, all seven parameters per scenario are elicited from a small language model under a fixed prompt protocol with worked anchor scenarios, repeated $k = 3$ times; the elicitation spread is propagated through every result. The pipeline is an experiment about LLM-based elicitation as much as a ranking.

2 Model

2.1 Structure

Latent state $\theta \in \{0, 1\}$, with the convention that $\theta = 1$ is the state in which responding is the correct action. The instrument produces a binary signal $x \in \{0, 1\}$. The agent then chooses an

action $a \in \{0, 1\}$ (respond / don't respond); the policy is derived by maximizing posterior expected utility, never elicited. Utilities are in the regret parameterization, $u(\theta, a=0) = 0$ for both states:

$$u(1, 1) = eB, \quad u(0, 1) = -K,$$

where $B > 0$ is the benefit of a correct response, $K > 0$ the net cost of a false-alarm response, and $e \in [0, 1]$ the action efficacy — the probability that the agent actually acts on the signal times the probability that the action delivers its benefit.

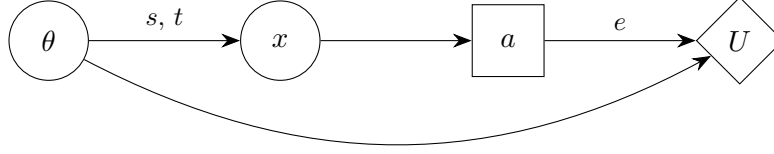


Figure 1: Influence diagram. The state θ (prior p) drives the signal x through sensitivity s and specificity t ; the agent observes x and chooses action a ; utility U depends on (θ, a) , with action efficacy e discounting the benefit edge.

2.2 Value of information in closed form

Write π for a belief $P(\theta=1)$. Acting optimally at belief π is worth

$$V(\pi) = \max(0, \pi eB - (1 - \pi)K),$$

the maximum over the two actions of posterior expected utility. With signal probabilities

$$P_1 = ps + (1 - p)(1 - t), \quad P_0 = 1 - P_1,$$

and posteriors $\pi_1 = ps/P_1$, $\pi_0 = p(1 - s)/P_0$ (a branch with zero probability contributes zero), the value of deciding *with* the signal is $P_1 V(\pi_1) + P_0 V(\pi_0)$, so

$$\text{EVSI} = P_1 V(\pi_1) + P_0 V(\pi_0) - V(p).$$

Perfect information replaces the signal with θ itself: the agent acts exactly when $\theta = 1$, which is worth $p eB + (1 - p) \cdot 0$. Hence

$$\text{EVPI} = p eB - V(p) = p eB - \max(0, p eB - (1 - p)K) = \min(p eB, (1 - p)K),$$

using $z - \max(0, z - w) = \min(z, w)$. The two arguments are the expected regrets of the two ignorant policies: never responding forgoes $p eB$; always responding costs $(1 - p)K$. Perfect information is worth exactly the smaller regret, and $0 \leq \text{EVSI} \leq \text{EVPI}$ always.

2.3 Per-scenario metrics

On every Monte Carlo draw we compute EVSI, EVPI, *efficiency* = EVSI/C (the primary ranking metric, ranked by median), *margin* = $\text{EVSI} - C$, and *headroom* = EVSI/EVPI where $\text{EVPI} > 0$ (how much of the perfect-information value this instrument delivers). Alongside the ranking we report $P_+ = P(\text{EVSI} > C)$ across draws: a high-efficiency scenario with low P_+ is fragile — its median hides a substantial probability that the measurement is not worth buying at all.

3 Method

3.1 Elicitation protocol

Each scenario’s seven parameters (p, s, t, e, B, K, C) are elicited from Claude Haiku via headless CLI calls (version 2.1.220 (Claude Code)), $k = 3$ independent repeats per scenario. Sampling temperature is not controllable from the CLI, so *repeats are the noise model*: cross-repeat spread is the empirical elicitation uncertainty. Calls run in an isolated context (no user configuration, no tools, a fixed system prompt), so the rendered prompt is the only input; its SHA-256 hash is stored with every response. The prompt (Appendix A) contains, in order: the model definition; two fully worked anchor scenarios with agreed numbers — an industrial vibration-survey decision and an everyday rapid strep test decision — to pin dollar and probability scales across scenarios; the scenario under elicitation; and per-parameter instructions requiring a short reasoning paragraph *before* the numbers. Each parameter is elicited as (p5, p50, p95) percentiles framed as “you would be surprised if the truth fell outside this interval.” Responses are validated (strict JSON, monotone percentiles, probabilities in $(0, 1)$, positive dollar amounts, an informativeness check $s_{50} > 1 - t_{50}$, non-degenerate prior); a failed attempt is retried once, then marked invalid. Validity across all attempts under the final protocol: 100% of 24 attempts. Total elicitation cost for the final protocol: \$1.27.

Scenarios themselves come from two sources: 8 hand-written seeds (doubling as smoke tests) and a proposer prompt that generates scenarios from ten seed domains under explicit constraints (one concrete decision-maker; θ a checkable binary fact; the instrument a purchasable measurement, not advice; stakes varied deliberately). No deduplication is applied: scenario text is its identity.

3.2 Percentiles to distributions

Dollar parameters (B, K, C) get lognormal fits; probability parameters (p, s, t, e) get Beta fits. Fits minimize squared quantile error against the three elicited percentiles (in log space for lognormals), initialized from moment matching. Every fit stores its parameters and residual in the database; badly asymmetric lognormal quantiles and Beta residuals above 0.02 raise a stored warning flag (7 warnings under the final protocol). Fits are never recomputed at analysis time.

3.3 Monte Carlo and sensitivity

Per scenario, 100 000 draws (seed 42) are taken with all seven parameters independent — a stated v1 assumption. Where a scenario has k valid repeats, each parameter draws from an equal-weight mixture over the k fitted distributions, so cross-repeat disagreement widens the intervals. Local sensitivity is the Spearman rank correlation between each parameter’s draws and the efficiency draws, per scenario. Global sensitivity aggregates $|\rho|$ over the top quartile of the ranking — the parameters that control the top of the list are where estimation effort should go next. Rank stability is computed from the aligned draws directly: the probability that a scenario ranks in the top 10 by efficiency across joint draws.

4 Scenario catalog

Table 1: Scenario catalog with pooled elicited medians (p50 across valid repeats). B, K, C in USD.

id	scenario	p	s	t	e	B	K	C
5	Soil pathogen assay before planting a high-value crop	0.15	0.85	0.92	0.75	50k	10k	280
1	Vibration survey before pulling a critical gearbox	0.10	0.82	0.90	0.82	150k	15k	1.2k
8	Counterparty credit report before extending 90-day trade credit	0.04	0.80	0.90	0.88	80k	20k	100
4	Standardized manipulation benchmark of a warehouse robot before a flee	0.15	0.85	0.91	0.80	250k	150k	10k
6	Structural inspection before purchasing a used mobile crane	0.10	0.88	0.92	0.82	50k	15k	3k
2	Pre-employment ergonomic screening before hiring a warehouse worker	0.12	0.75	0.85	0.65	15k	5k	250
7	Sepsis-risk panel for a newly admitted patient	0.05	0.82	0.78	0.80	25k	1k	150
3	Dangerous-capability evaluation of a frontier AI model before deployment	0.18	0.75	0.80	0.75	5M	1M	150k

5 Results

Table 2: Final ranking by median efficiency (EVSI/C). P_+ = $P(\text{EVSI} > C)$ across draws; last column = parameter with largest $|\rho|$ vs efficiency.

rank	id	scenario	eff_{q05}	eff_{q50}	eff_{q95}	P_+	top
1	5	Soil pathogen assay before planting a high-value crop	0	7.35	58.4	0.81	C
2	1	Vibration survey before pulling a critical gearbox	0	2	18.6	0.62	C
3	8	Counterparty credit report before extending 90-day trade credit	0	0.789	97.5	0.49	B
4	4	Standardized manipulation benchmark of a warehouse robot before	0	0.597	9.76	0.42	B
5	6	Structural inspection before purchasing a used mobile crane	0	0.529	4.66	0.34	B
6	2	Pre-employment ergonomic screening before hiring a warehouse wor	0	0.478	8.85	0.42	B
7	7	Sepsis-risk panel for a newly admitted patient	0	0	10.1	0.32	t

rank	id	scenario	eff _{q05}	eff _{q50}	eff _{q95}	P_+	top
8	3	Dangerous-capability evaluation of a frontier AI model before de	0	0	12.9	0.25	s

5.1 Reading the top of the ranking

(To be written after the final protocol run.)

6 Sensitivity and elicitation quality

Global sensitivity over the top-quartile scenarios puts $|\rho|$ per parameter at: p : 0.19, s : 0.18, t : 0.16, e : 0.04, B : 0.32, K : 0.18, C : 0.36 — the ranking’s top is controlled chiefly by C , B , p . Cross-repeat noise (median over scenarios of the relative p50 spread) is: p : 0.53, s : 0.04, t : 0.03, e : 0.08, B : 0.89, K : 1.18, C : 0.72.

6.1 Protocol comparison

Spearman rank correlations of median efficiency between the latest runs of each protocol (rows/columns are protocols; the manual column is the hand-entered seed baseline over its 8 scenarios):

	p000_manual	p001
p000_manual	–	0.31
p001	0.31	–

(To be written after the protocol iterations.)

7 Limitations

Per-measurement value, not business value. With value capture and buyer counts removed by design, the ranking measures how much one observation is worth to one decision-maker per dollar it costs to produce. A top-ranked scenario is not a business case: it says nothing about how many such decisions exist, whether the decision-maker can be reached, or what fraction of the information value a seller could charge.

Independence. All seven parameters are drawn independently. In reality, e.g., high-stakes scenarios (B large) plausibly come with more expensive instruments (C large) and more careful priors; ignoring the correlations mostly widens intervals but can also bias efficiency medians.

The efficacy asymmetry. e multiplies only the benefit branch: $V(\pi) = \max(0, \pi e B - (1 - \pi)K)$. Since e includes the probability that the agent acts at all, a fully symmetric treatment would discount the false-alarm cost by the acting probability too; v1 keeps the simpler, conservative form (it understates the value of acting for noncompliant agents).

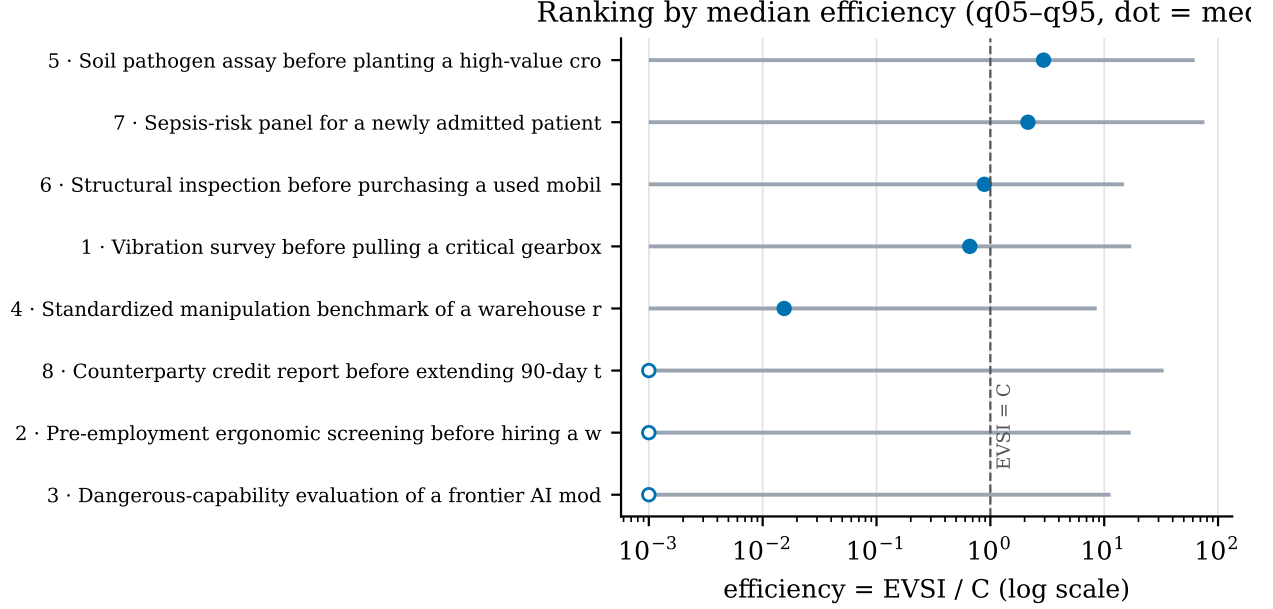


Figure 2: Top 25 scenarios by median efficiency $EVSI/C$. Intervals are q05–q95 across Monte Carlo draws; dots are medians (open dots: median 0, shown at the 10^{-3} axis floor). The dashed line marks $EVSI = C$: scenarios to its right buy more than a dollar of decision value per dollar of measurement.

A small model as elicitor. Haiku is cheap enough to elicit hundreds of scenario-repeats, but small-model calibration is a real risk. The mitigations are the two anchors (scale pinning) and k repeats (noise measurement); Section 6 shows the resulting noise levels per parameter, and the protocol-comparison matrix shows how much the ranking moves when the prompt changes. Both are evidence about — not proof of — elicitation quality. The hand-entered seed baseline correlates with the final protocol’s ranking at $\rho = 0.31$ over 8 shared scenarios.

Selection effects of validation. The informativeness check ($s_{50} > 1 - t_{50}$) rejects elicitations rather than repairing them; a scenario whose instrument haiku consistently sees as uninformative drops out of the ranking instead of appearing with $EVSI \approx 0$.

Deferred by design. Capture fraction, number of buyers, annualization and recurrence, rivalry, compliance or stamp demand, risk aversion, parameter correlations, retrodiction validation, richer state and signal spaces (POMDPs), and deduplication are all out of scope for v1.

8 Learnings

(To be written after the protocol iterations.)

A Elicitation prompt template

The exact template of the final protocol (placeholders `$title`, `$agent`, `$decision`, `$theta.definition`, `$instrument` are substituted per scenario):

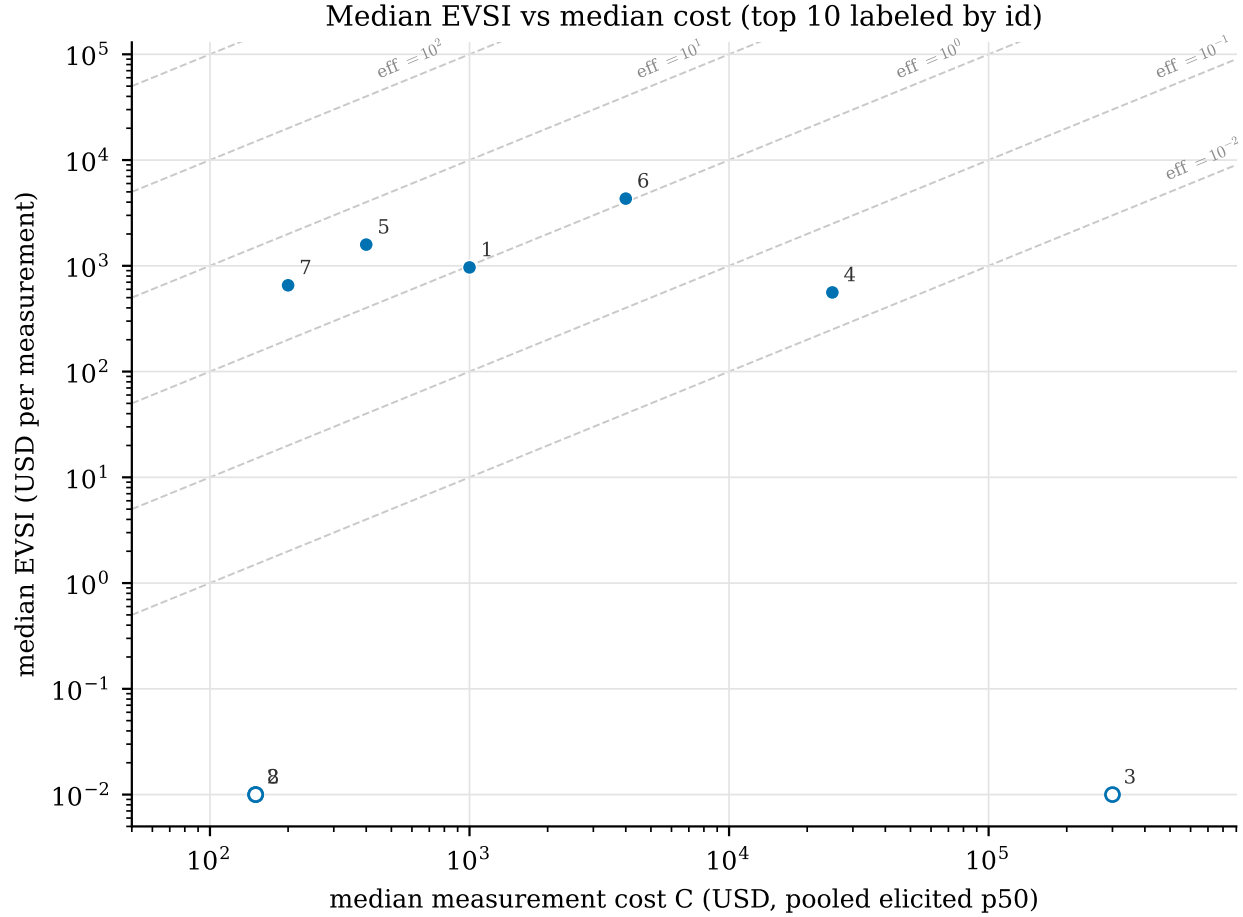


Figure 3: Median EVSI against median measurement cost, log-log, with iso-efficiency diagonals. The top 10 scenarios by median efficiency are labeled by catalog id. Open dots at the bottom edge are scenarios whose median EVSI is 0.

```

You are performing a structured quantitative elicitation of the seven parameters of a
canonical Bayesian value-of-information model, for one specific scenario.

## The model

A single agent faces a single binary decision: respond (a=1) or do not respond (a=0).
The world is in a hidden binary state theta. Convention: theta=1 is the state in which
responding is the correct action.
The agent can buy exactly one measurement from an instrument. It produces a binary signal x,
where x=1 suggests theta=1.
The seven parameters:
- p: prior probability P(theta=1), given everything the agent already knows WITHOUT buying
this measurement. Base rates, visible symptoms, free records, common knowledge: all of that
is already inside p.
- s: sensitivity P(x=1 | theta=1) - the probability the instrument fires when the condition
is truly present.
- t: specificity P(x=0 | theta=0) - the probability the instrument stays silent when the
condition is truly absent.
- e: action efficacy = P(the agent actually responds after a positive signal) x P(the
response achieves its benefit). A number in [0,1].
- B: benefit of a correct response when theta=1, in USD, for this one decision. B = u(1,1) -
u(1,0): what responding gains over not responding when the condition is present.
- K: net cost of a false-alarm response when theta=0, in USD, for this one decision. K = u

```

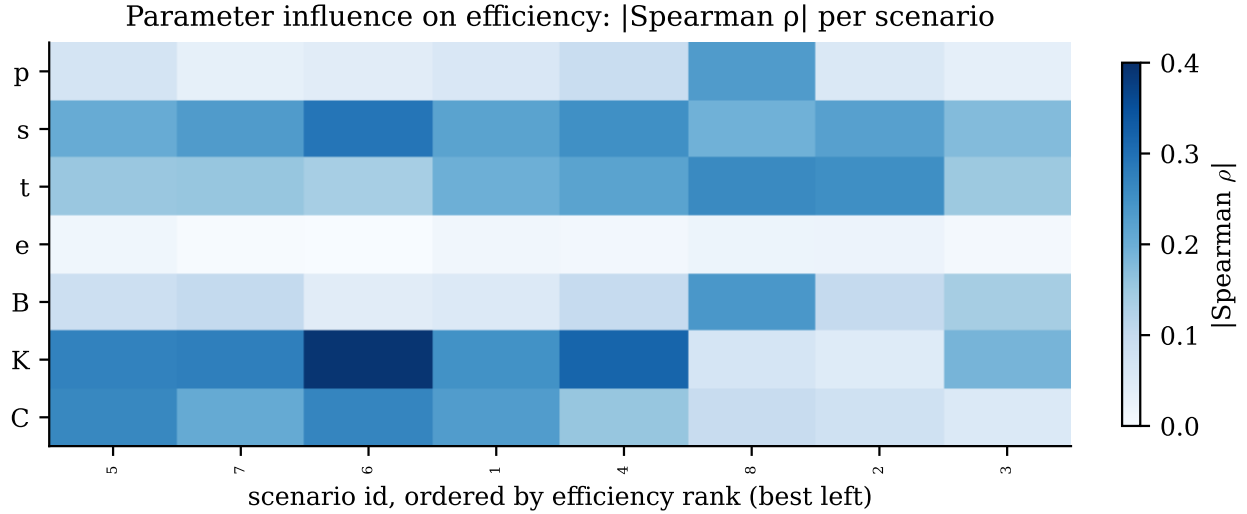


Figure 4: Absolute Spearman correlation between each parameter's draws and the efficiency draws, per scenario, scenarios ordered by rank.

```
(0,0) - u(0,1): what responding loses over not responding when the condition is absent.
- C: cost to produce and deliver ONE measurement, in USD.

All USD amounts are per single decision. Never use market sizes, annual totals, portfolio-
wide or fleet-wide numbers.

## Anchor scenarios (calibrate your scales against these agreed numbers)

Anchor A1. A plant reliability engineer decides whether to pull a critical gearbox for
overhaul. theta=1 = an incipient bearing fault is present. Instrument = one third-party
vibration-analysis survey.
- p: 0.02 / 0.08 / 0.25
- s: 0.60 / 0.80 / 0.95
- t: 0.70 / 0.90 / 0.98
- e: 0.50 / 0.80 / 0.95
- B (avoided unplanned failure minus planned repair): 20,000 / 150,000 / 1,500,000 USD
- K (unnecessary teardown + downtime): 2,000 / 10,000 / 60,000 USD
- C: 300 / 1,000 / 5,000 USD

Anchor A2. A primary-care physician decides whether to start a patient with acute sore
throat on antibiotics. theta=1 = group A streptococcal infection is present. Instrument =
one rapid antigen strep test.
- p: 0.05 / 0.15 / 0.35
- s: 0.70 / 0.85 / 0.95
- t: 0.90 / 0.95 / 0.99
- e (patient adheres x antibiotics deliver their benefit): 0.60 / 0.85 / 0.98
- B (avoided complications, shortened illness, reduced transmission): 50 / 300 / 3,000 USD
- K (unnecessary antibiotics: side effects, resistance contribution per decision): 20 / 100
/ 1,000 USD
- C: 5 / 15 / 50 USD

Each triple is p5 / p50 / p95 of the elicitor's belief. Note the scales: A1 is an industrial
six-figure-stakes decision, A2 is an everyday hundreds-of-USD decision. Place the scenario
below on a consistent scale relative to both.

## Scenario to elicit

- Title: $title
- Agent (the decision-maker): $agent
- Decision (respond vs not): $decision
- State definition: $theta_definition
```

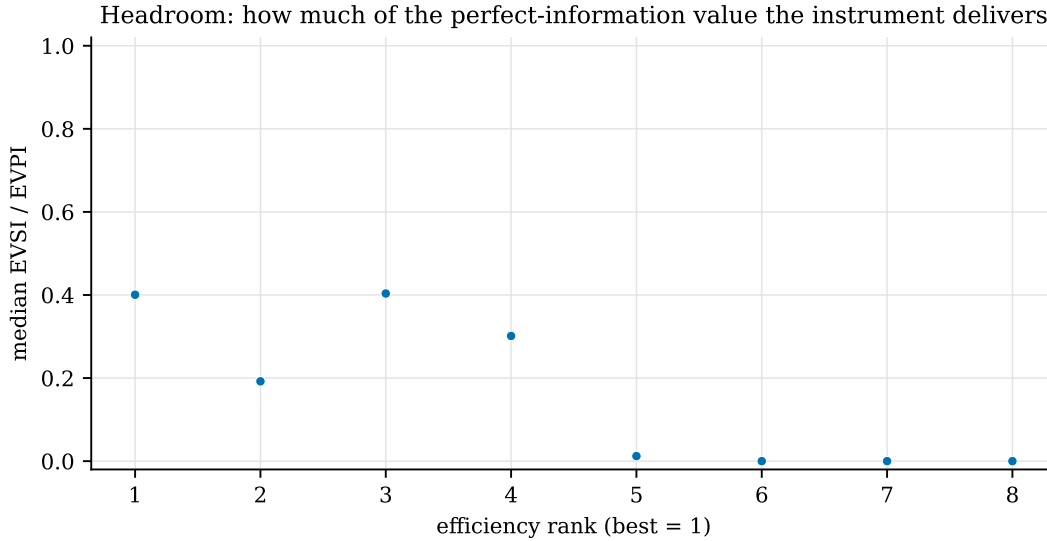


Figure 5: Median headroom EVSI/EVPI against rank. Low headroom at high rank marks decisions where a better instrument than the current one is worth building.

```
- Instrument (the purchasable measurement): $instrument

## Instructions

For each of the seven parameters, in the order p, s, t, e, B, K, C:
1. First write a short paragraph of reasoning (2-4 sentences): what drives this quantity for
   THIS scenario, referencing base rates or typical magnitudes where you can.
2. Then commit to three percentiles p5, p50, p95. Read them as: p50 is your median estimate;
   you would be genuinely surprised if the true value fell outside [p5, p95]. They must be
   strictly increasing: p5 < p50 < p95.
3. State the unit ("probability" or "USD").

Remember for p: it must already reflect all information the agent has without buying this
measurement. If a free substitute for the instrument exists, its effect lives inside p, not
inside s or t.
Remember for B, K, C: single-decision USD amounts, consistent in scale with the anchors.

## Output

Output strict JSON only. No markdown fences, no prose outside the JSON. Exactly this shape,
with all seven parameters filled in:

{
  "parameters": {
    "p": {"reasoning": "...", "p5": 0.0, "p50": 0.0, "p95": 0.0, "unit": "probability"},
    "s": {"reasoning": "...", "p5": 0.0, "p50": 0.0, "p95": 0.0, "unit": "probability"},
    "t": {"reasoning": "...", "p5": 0.0, "p50": 0.0, "p95": 0.0, "unit": "probability"},
    "e": {"reasoning": "...", "p5": 0.0, "p50": 0.0, "p95": 0.0, "unit": "probability"},
    "B": {"reasoning": "...", "p5": 0.0, "p50": 0.0, "p95": 0.0, "unit": "USD"},
    "K": {"reasoning": "...", "p5": 0.0, "p50": 0.0, "p95": 0.0, "unit": "USD"},
    "C": {"reasoning": "...", "p5": 0.0, "p50": 0.0, "p95": 0.0, "unit": "USD"}
  }
}
```

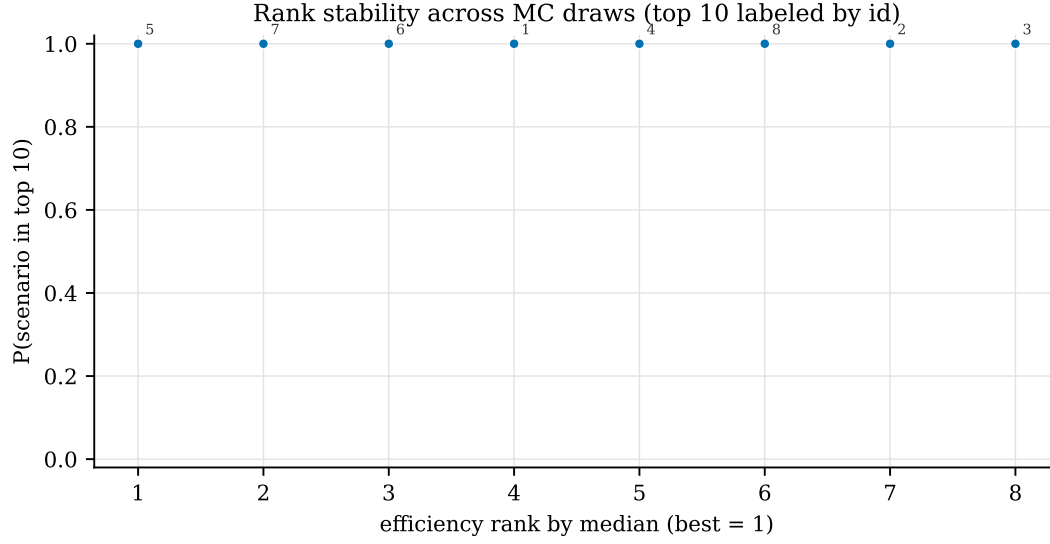


Figure 6: Rank stability: probability of membership in the top 10 by efficiency, across aligned Monte Carlo draws.

B Reproduction

Everything downstream of the elicitation database is deterministic given the stored seed (run 3: seed 42, 100 000 draws, code revision 7c4f42846081). The database stores every prompt hash, raw model response, fitted distribution, and run configuration; figures and tables in this report are generated exclusively from it. See `README.md` in the repository root for the run order.

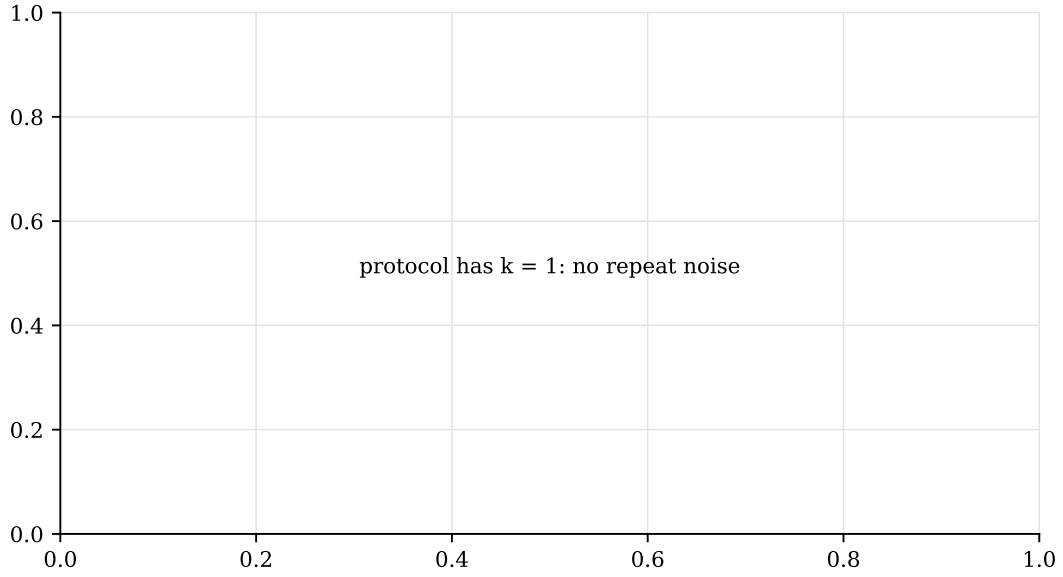


Figure 7: Cross-repeat elicitation noise: spread of the median estimate across $k = 3$ repeats, relative to the pooled median, per parameter.

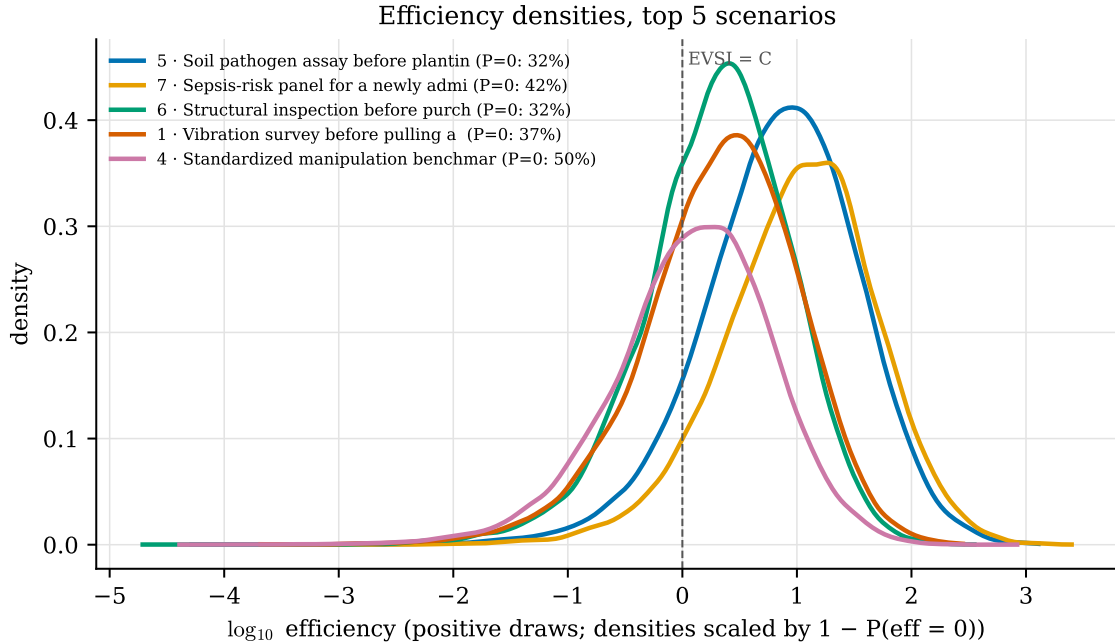


Figure 8: Efficiency densities of the top 5 scenarios over positive draws ($\log_{10}$ axis), scaled by $1 - P(\text{eff} = 0)$; the point mass at zero is the share of draws in which the signal never changes the decision.